

Pandas Practice Set: Cleaning, Transformation, and Reshaping

Data Cleaning: Handling Missing Values & Duplicates

1. How do you check the total number of missing (null) values in each column of a pandas DataFrame?
2. What is the difference between calling `df.dropna()` and `df.dropna(axis=1)` ?
3. Write a pandas statement to fill all missing values in an "Age" column with the mean of that column.
4. Explain the difference between `df.fillna(method="ffill")` and `df.fillna(method="bfill")` .
5. What method is used to identify duplicate rows in a DataFrame, and how do you remove them entirely?
6. How do you drop duplicates based only on a specific subset of columns, for instance, "Name" and "Age" ?

String Manipulations

7. Write the code to convert all text in the "Name" column of a DataFrame to lowercase.
8. How can you check if the "City" column contains the substring "delhi" , while making the search case-insensitive?
9. Given an "Email" column containing complete email addresses, how would you split the strings into two parts at the "@" symbol?

Data Types & Feature Engineering

10. How do you convert the data type of an "Age" column from float or string to integer?
11. What insights does the `df.info()` method provide about a DataFrame?
12. Why is it recommended to use `.copy()` when assigning a filtered DataFrame to a new variable (e.g., `df2 = df.dropna().copy()`)?
13. Write the code to create a new column "Age Group" using a lambda function with `apply()` , where anyone with an Age ≥ 25 is marked as "Adult" and others as "Minor" .
14. If you have a custom function `def categorize_age(x): ...` , how do you apply it to the "Age" column to create a new column?
15. How can you systematically change values in a "Gender" column from "M" , "F" , "O" to "Male" , "Female" , "Other" using the `.map()` method?
16. How do you use the `.replace()` method to change the value "Delhi" to "New Delhi" and "Mumbai" to "New Mumbai" in the "City" column?

Data Transformation & Sorting

17. Write the code to sort a DataFrame by the "Age" column in descending order.
18. How can you sort a DataFrame based on multiple columns, for example, first by "City" and then by "Age" ?
19. After sorting or filtering a DataFrame, the index can become out of order. What method and parameters do you use to reset the index to standard sequential integers and drop the old index?
20. What is the purpose of the `df.sort_index()` method?
21. Write the code to assign a rank to rows based on the "Age" column in descending order using the 'dense' ranking method.
22. If your DataFrame has multiple columns, but you only want to keep and reorder specific ones ("Name" , "Age" , "City"), how would you achieve this?

DataFrame Creation & Reshaping (Melt)

23. Write a brief code snippet to create a pandas DataFrame from a standard Python dictionary.
24. Given a DataFrame with horizontal columns: "Name" , "Math" , "Science" , and "English" , write the code using `melt()` to unpivot this data. The output should keep "Name" as the identifier, combine the subjects into a new column called "Subject" , and place their corresponding values in a column called "Score" .
25. Given the following melted DataFrame with columns "Name" , "Subject" , and "Score" , write the code using `pivot()` to convert it back to the original wide format where each subject ("Math" , "Science" , "English") becomes a separate column and "Name" remains as the index.